

# Untitled

June 2, 2025

## Parte 1

```
[9]: import pandas as pd

# Load the CSV
df = pd.read_csv("data/healthcare_dataset-20250506.csv", sep=';')

# Show the first few rows
print(df.head())

# Show data types of each column
print(df.dtypes)
```

|   | Name          | Age | Gender | Blood Type | Medical Condition | Date of Admission | \ |
|---|---------------|-----|--------|------------|-------------------|-------------------|---|
| 0 | Bobby JacksOn | 30  | Male   | B-         | Cancer            | 31/01/2024        |   |
| 1 | LesLie TErRy  | 62  | Male   | A+         | Obesity           | 20/08/2019        |   |
| 2 | DaNnY sMitH   | 76  | Female | A-         | Obesity           | 22/09/2022        |   |
| 3 | andrEw waTtS  | 28  | Female | O+         | Diabetes          | 18/11/2020        |   |
| 4 | adriENNE bEll | 43  | Female | AB+        | Cancer            | 19/09/2022        |   |

|   | Doctor           | Hospital                   | Insurance Provider | \ |
|---|------------------|----------------------------|--------------------|---|
| 0 | Matthew Smith    | Sons and Miller            | Blue Cross         |   |
| 1 | Samantha Davies  | Kim Inc                    | Medicare           |   |
| 2 | Tiffany Mitchell | Cook PLC                   | Aetna              |   |
| 3 | Kevin Wells      | Hernandez Rogers and Vang, | Medicare           |   |
| 4 | Kathleen Hanna   | White-White                | Aetna              |   |

|   | Billing Amount | Room Number | Admission Type | Discharge Date | Medication  | \ |
|---|----------------|-------------|----------------|----------------|-------------|---|
| 0 | 18856.281306   | 328         | Urgent         | 02/02/2024     | Paracetamol |   |
| 1 | 33643.327287   | 265         | Emergency      | 26/08/2019     | Ibuprofen   |   |
| 2 | 27955.096079   | 205         | Emergency      | 07/10/2022     | Aspirin     |   |
| 3 | 37909.782410   | 450         | Elective       | 18/12/2020     | Ibuprofen   |   |
| 4 | 14238.317814   | 458         | Urgent         | 09/10/2022     | Penicillin  |   |

|   | Test Results |
|---|--------------|
| 0 | Normal       |
| 1 | Inconclusive |
| 2 | Normal       |

```

3      Abnormal
4      Abnormal
Name          object
Age           int64
Gender        object
Blood Type    object
Medical Condition object
Date of Admission object
Doctor        object
Hospital      object
Insurance Provider object
Billing Amount float64
Room Number   int64
Admission Type object
Discharge Date object
Medication    object
Test Results  object
dtype: object

```

```
[2]: !pip install pymongo
```

```

Collecting pymongo
  Downloading pymongo-4.13.0-cp311-cp311-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (22 kB)
Collecting dnspython<3.0.0,>=1.16.0 (from pymongo)
  Downloading dnspython-2.7.0-py3-none-any.whl.metadata (5.8 kB)
Downloading
pymongo-4.13.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.4
MB)
1.4/1.4 MB
9.3 MB/s eta 0:00:00:00:0100:01
Downloading dnspython-2.7.0-py3-none-any.whl (313 kB)
313.6/313.6 kB
9.9 MB/s eta 0:00:00
Installing collected packages: dnspython, pymongo
Successfully installed dnspython-2.7.0 pymongo-4.13.0

```

Parte 2

```

[5]: from pymongo import MongoClient
import os

MONGO_USER = os.getenv("MONGO_INITDB_ROOT_USERNAME", "root")
MONGO_PASS = os.getenv("MONGO_INITDB_ROOT_PASSWORD", "password")
MONGO_HOST = os.getenv("MONGO_HOST", "mongo")
MONGO_DB = "healthcare_dataset"

mongo_uri = f"mongodb://{MONGO_USER}:{MONGO_PASS}@{MONGO_HOST}:27017/"

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```

client = MongoClient(mongo_uri)
db = client[MONGO_DB]
count = db.patients.count_documents({})
print("1 Total patients in collection:", count)

```

Total patients in collection: 55500

```

[18]: from datetime import datetime
      from pymongo import MongoClient
      import os

      MONGO_USER = os.getenv("MONGO_INITDB_ROOT_USERNAME", "root")
      MONGO_PASS = os.getenv("MONGO_INITDB_ROOT_PASSWORD", "password")
      MONGO_HOST = os.getenv("MONGO_HOST", "mongo")
      MONGO_DB = "healthcare_dataset"

      mongo_uri = f"mongodb://{MONGO_USER}:{MONGO_PASS}@{MONGO_HOST}:27017/"
      client = MongoClient(mongo_uri)
      db = client[MONGO_DB]

      cutoff = datetime(2023, 1, 1)

      # Limit to first 10 results
      cursor = db.patients.find({"date_of_admission": {"$gt": cutoff}}).limit(10)

      for doc in cursor:
          name = doc.get("name", "")
          doa = doc.get("date_of_admission")
          if doa:
              print(f"{name:30s} admitted {doa.date()}")

```

|                |                     |
|----------------|---------------------|
| Bobby JacksOn  | admitted 2024-01-31 |
| EMILY JOHNSOn  | admitted 2023-12-20 |
| aaRon MARTiNeZ | admitted 2023-08-13 |
| tIMOTHY burNs  | admitted 2023-06-28 |
| cathy sMaLl    | admitted 2023-12-23 |
| jOSHUA OLiVer  | admitted 2023-10-03 |
| WILLIAM cOOPEr | admitted 2023-05-18 |
| Erin oRTEga    | admitted 2023-05-24 |
| kyLE bEnneTT   | admitted 2023-09-09 |
| mIchael LiU    | admitted 2024-04-05 |

```

[19]: count_over_50 = db.patients.count_documents({"age": {"$gt": 50}})
      print("Patients older than 50:", count_over_50)

```

Patients older than 50: 28667

```
[20]: thomas_count = db.patients.count_documents({
        "name": {"$regex": r"^Thomas\s", "$options": "i"}
    })
    print("3 Patients with first name 'Thomas':", thomas_count)
```

3 Patients with first name 'Thomas': 397

```
[21]: pipeline = [
        {"$group": {"_id": "$medical_condition", "count": {"$sum": 1}}},
        {"$sort": {"_id": 1}}
    ]
    results = list(db.patients.aggregate(pipeline))

    print("3 Count per distinct medical condition:")
    for r in results:
        print(f"    {r['_id']:15s}: {r['count']}")
```

3 Count per distinct medical condition:

|              |        |
|--------------|--------|
| Arthritis    | : 9308 |
| Asthma       | : 9185 |
| Cancer       | : 9227 |
| Diabetes     | : 9304 |
| Hypertension | : 9245 |
| Obesity      | : 9231 |

```
[22]: pipeline = [
        {"$group": {"_id": "$medication", "count": {"$sum": 1}}},
        {"$sort": {"_id": 1}}
    ]
    results = list(db.patients.aggregate(pipeline))

    print("3 Medication usage counts:")
    for r in results:
        print(f"    {r['_id']:12s}: {r['count']}")
```

3 Medication usage counts:

|             |         |
|-------------|---------|
| Aspirin     | : 11094 |
| Ibuprofen   | : 11127 |
| Lipitor     | : 11140 |
| Paracetamol | : 11071 |
| Penicillin  | : 11068 |

```
[23]: lipitor_cursor = db.patients.find({"medication": "Lipitor"})

lipitor_list = list(lipitor_cursor)
print("4 ")
print(f"Total Lipitor patients: {len(lipitor_list)}\n")
```

```
4
Total Lipitor patients: 11140
```

```
[24]: print("5 Sample Lipitor patients (first 5):")
for doc in lipitor_list[:5]:
    name = doc.get("name", "")
    age = doc.get("age", "")
    condition = doc.get("medical_condition", "")
    print(f" - {name}, age {age}, condition: {condition}")
```

```
5 Sample Lipitor patients (first 5):
- aaRon MARTiNeZ, age 38, condition: Hypertension
- rObErT bAuer, age 68, condition: Asthma
- ChRISToPHEr BRiGhT, age 48, condition: Asthma
- KathRYn StewArt, age 58, condition: Arthritis
- dR. EilEEen thomPsoN, age 59, condition: Asthma
```